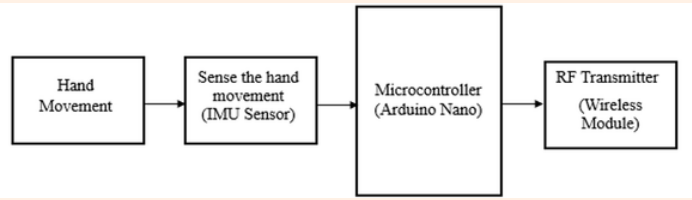


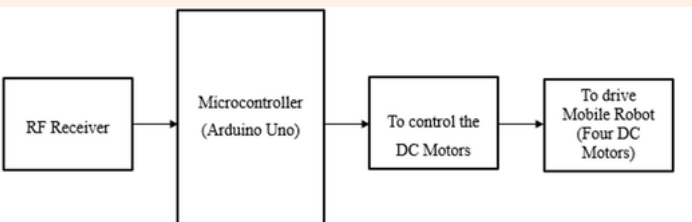
AAIM AND OBJECTIVES

The main aim of this system is to develop and control the IMU sensor, connect with wireless communication, and control the mobile robot's movements accurately in real time based on the hand gestures.

- To develop and implement a wireless communication system to transmit gesture commands to the mobile robot.
- To control a mobile robot based on recognized hand gesture such as forward, backward, left, right, orientation and stop with hand gesture.
- To improve system accuracy and reliability through proper calibration and filtering of IMU sensor data.

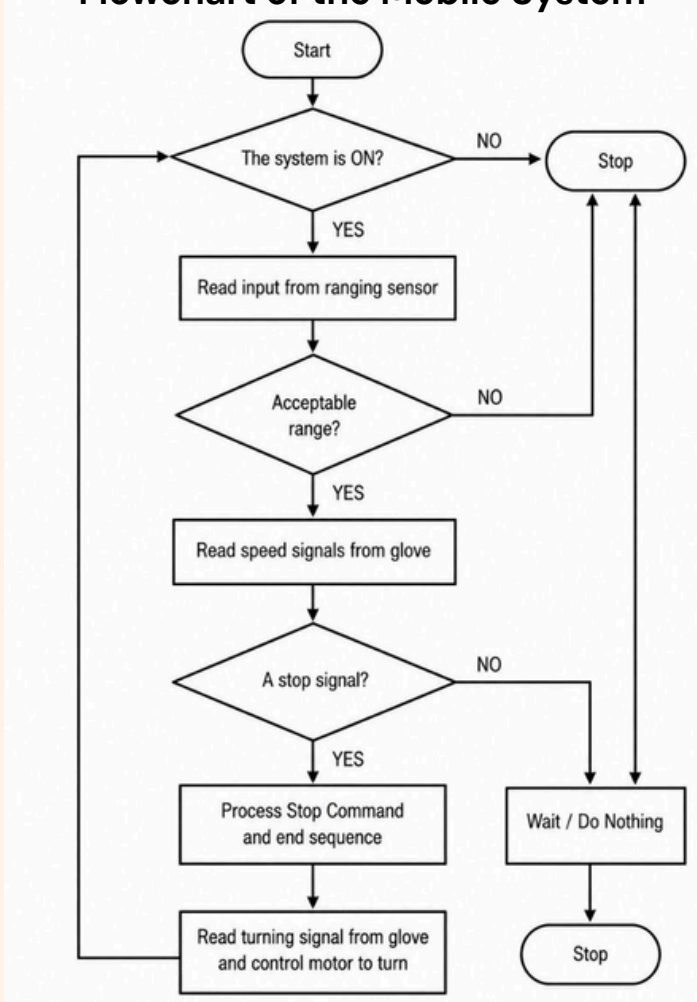


Block Diagram of the Hand Gesture Control System

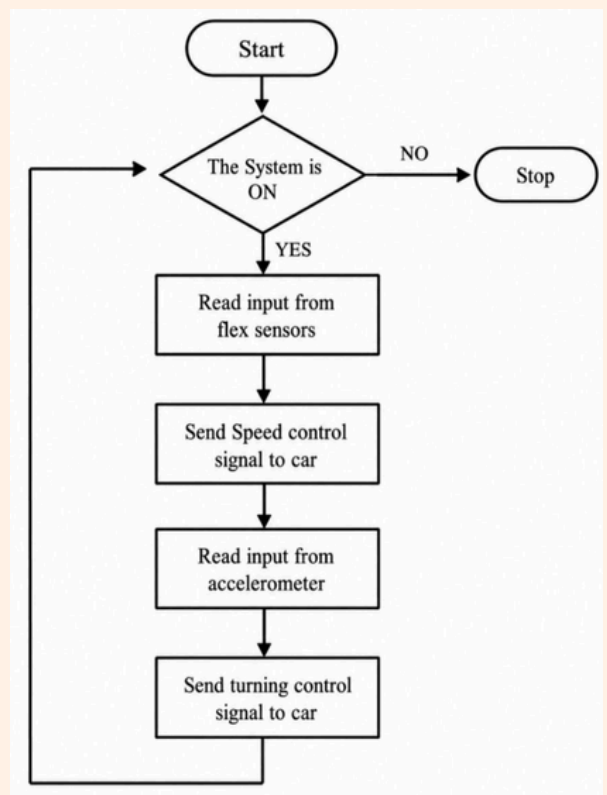


Block Diagram of the Mobile Robot

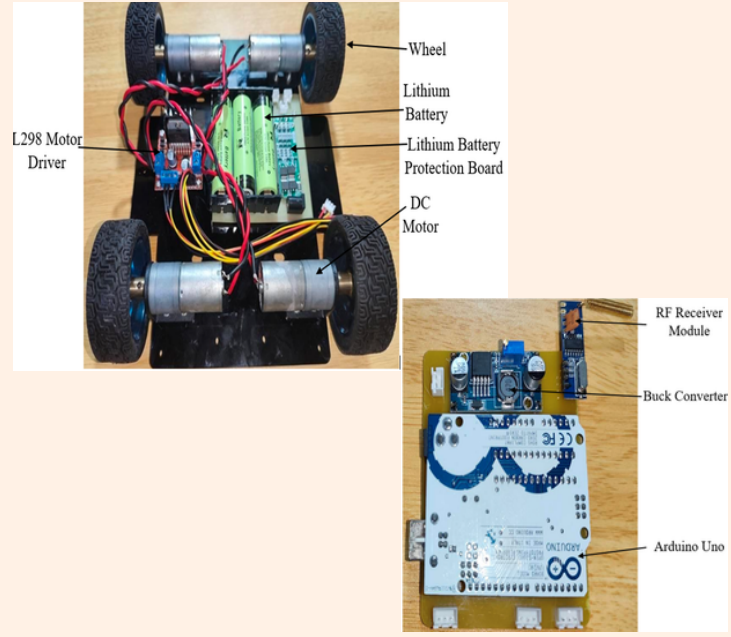
Flowchart of Receiver Section Flowchart of the Mobile System



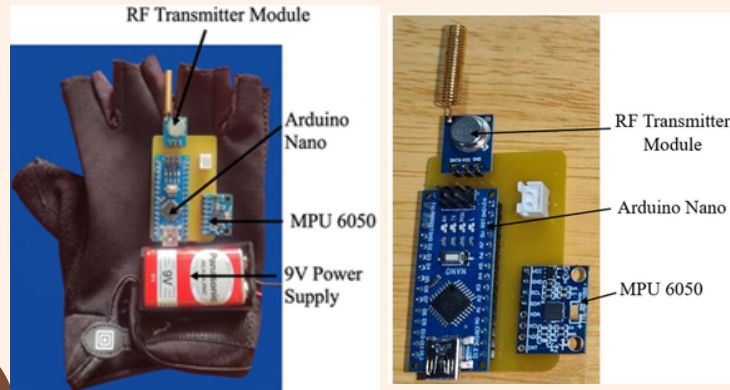
Transmitter Section Flowchart



Hardwre Designation



PROTOTYPE



CONCLUSION

- The proposed system successfully demonstrates wireless control of a mobile robot using hand gestures.
- Experimental results show that the forward, backward, left, right and rotation conditions can perform gesture-based navigation with acceptable accuracy.
- The integration of Arduino's, RF modules, and motor drivers enables reliable real-time robot movement.

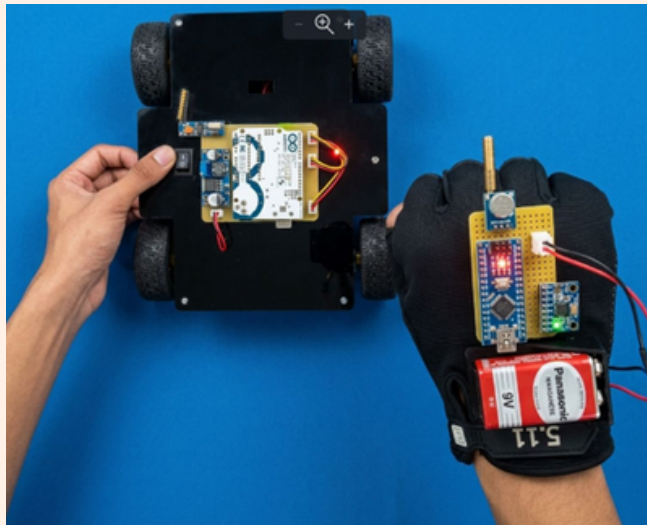


Applied Areas

- Industrial Material Handling
- Search and Rescue Missions
- Healthcare Assistance
- Education and Research Projects

SYSTEM PROCESS OF THE PROPOSED SYSTEM

- Step (1) : The system starts.
- Step (2) : The user moves their hand.
- Step (3) : The system reads the hand movement using IMU-sensors.
- Step (4) : The system checks whether the hand moves up.
- Step (5) : If yes, the car moves backward.
- Step (6) : If not, the system checks whether the hand moves down.
- Step (7) : If yes, the car moves forward.
- Step (8) : If not, the system checks whether the hand moves right.
- Step (9) : If yes, the car moves right.
- Step (10) : If yes, the car moves left.
- Step (11) : If none of these movements are detected and the hand is in a stable position,
- Step (12) : The car remains at rest (stop condition).
- Step (13) : The process continues repeatedly while the system is powered on.

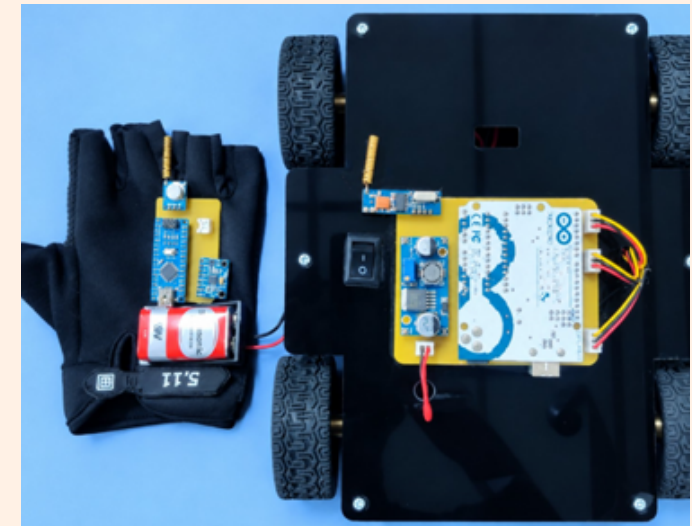


PROJECT SHOW & JOB FAIR 2026



TECHNOLOGICAL UNIVERSITY (MANDALAY)

DEPARTMENT OF MECHATRONIC ENGINEERING



WIRELESS CONTROL OF A MOBILE ROBOT USING IMU- BASED HAND GESTURE

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